MILITARY SPECIFICATION

TRAINING AID, ENLARGED RADIO MAGNETIC INDICATOR

This specification has been approved by the Naval Training Device Center, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for the production of an Enlarged Radio Magnetic Indicator for use as a training aid.

1.1.1 This item has been designated by the Naval Training Device Center as Device 2B25/1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner

Military

MIL-P-116	Preservation; Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible
MIL-T-23991	Training devices, Military; General Specification For

FSC 6910

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. Requirements

3.1 Materials.- Material used shall be as specified in 3.1.1 through 3.1.4 of this specification. Materials not specified shall be in accordance with MIL-T-23991.

3.1.1 Lacquers, inks or dyes.- Lacquers, inks or dyes shall form a permanent bond with the surface applied to and shall not cause or develop a crazed surface.

3.1.2 Metal parts and fastenings.- All metal parts and fastenings shall be of corrosion resisting steel, or be satin-finish chromium-plated or anodized clear, as applicable.

3.1.3 Plastic.- Plastics shall be durable and resistant to cracking, brittleness, discoloration and deformation upon aging, or when subjected to the range of temperatures specified herein. Transparent panels shall be made of rigid vinyl or acrylic.

3.1.4 Hardboard.- Hardboard shall be of the tempered variety, smooth on both sides, hard surfaced and moisture resistant for a paint finish. U. S. Gypsum "Duron" or material equivalent in strength, rigidity and smoothness of surface is acceptable.

3.2 Design.- The training aid shall represent a seven times enlarged version of the hand held device 2B25. It shall serve as a classroom demonstration aid for use by the instructor. The general arrangement and concept is shown in Figure 1. The compass rose and the two indicators shall rotate about a centerpost. A felt spacer shall be between each rotating part and the friction shall be controllable by a hand operated, self-locking nut. In order to prevent rotating parts from touching and scraping, judiciously placed Teflon buttons shall be used.

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3.3 Reliability.- The training aid shall be constructed to withstand use in classrooms for five hours per week for five years. The training aid shall not be adversely affected in any manner during storage and transportation when under ambient temperature ranging from minus 5 to plus 125 degrees Fahrenheit and relative humidity ranging to 100 percent. The training aid shall be assembled in such a manner that no fixed part or assembly shall become undesirably free or sluggish in operation and no movable part shall bind or seize. No degradation shall be caused in the mechanical performance stipulated for the training aid because of shipment, operation or storage.

3.4 Details of components.- The training aid shall consist of the following major components:

(a) Baseplate

(b) Compass rose

(c) Two indicators

(d) Centerpost.

3.4.1 Baseplate.- The baseplate shall be made from 3/16 inch hardboard, shall be square, 36 inches on edge and shall have rounded corners with a two inch radius. The front face shall represent the base of Device 2B25, a white disc with black markings. The area outside the disc, and the back of the baseplate shall be painted gray, No. 26231 of FED-STD-595. Two hangers shall be provided at the upper edge of the baseplate.

3.4.1.1 An emblem and product markings shall be imprinted in black on a white background as shown in Fig. 1.

3.4.2 Compass rose.- The compass rose shall be made from 1/8 inch hardboard. It shall be white on both sides, with black markings on the front face. The markings shall be distinct and bold so that they can be read from 25 feet distance. Twelve one inch holes, equally spaced on a nine (9) inch radius shall be provided to assist in turning the disc.

3.4.3 Two indicators.- The two indicators shall be printed in black on two clear, transparent plastic plates. Each plate shall be 25 by 8 by 1/8 inch.

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3.4.4 Centerpost.- The centerpost general arrangement and concept is shown in Figure 2. The post and selflocking knurled nut shall be made of corrosion resistant steel or brass with satin-finish chrome plate. The centerpost shall be permanently fastened to the baseplate. Felt washers, approximately 2 inches in diameter by 1/8 inch shall be on each side of a rotating part.

3.5 Construction.- The training aid shall be constructed from hardboard and plastic components painted, printed or silk screened in accordance with 3.2 and 3.4. All markings shall be protected by a clear transparent coat of lacquer or fused on plastic heavy enough to satisfy the requirements under 3.3.

3.6 Dimensions.- The overall dimensions of the training aid shall be 36 by 36 by 2 inches.

3.7 Color and finish.- Colors shall be as specified under 3.4. All finished surfaces shall be free from toolmarks, scratches and sharp edges.

3.8 Nameplates and product markings.- Nameplates and product markings will be specified by the procuring activity and shall be permanently affixed to the training aid for identification. (See 3.4.1.1)

3.9 Government-furnished property.- Government-furnished property will be specified by the procuring activity.

3.10 Workmanship.- The training aid shall be constructed and finished with particular attention given to neatness, sturdiness, correctness of detail, freedom from sharp edges and burrs, ease of operation and clarity, sharpness and accuracy of graduations and lettering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- Inspections to be performed are classified as follows:

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(a) In-process inspection

(b) Quality conformance inspection.

4.2.1 In-process inspection.- The in-process inspection shall include such visual and mechanical examinations and testing of materials, sub-assemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.2.2 Quality conformance inspection.- The quality conformance inspection shall consist of a preproduction inspection and an inspection of the production units.

4.2.2.1 A preproduction inspection shall be conducted on a preproduction unit and shall consist of all the examinations and tests specified in 4.3 and 4.4.

4.2.2.2 Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-approved pre-production unit.

4.3 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Details of components

(d) Dimensions

(e) Color and finish

(f) Nameplates and product markings

(g) Workmanship.

4.4 Tests.- Each training aid shall be subjected to a functional test consisting of the following:

(a) Move each movable component through the full range of movement provided for in the design

(b) Check that friction can be adequately controlled as required to make each component stop and stay in any position where placed.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring activity.

5.1.1 Level A.- For Level A, each training aid shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring activity.

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2, or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water.

5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid, Radio Magnetic Indicator, is intended for classroom use in the instruction of students in a course of aerial navigation.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions.- Except as indicated in 6.3.1 and 6.3.2, definitions will be furnished upon written request to the procuring activity.

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6.3.1 Preproduction unit.- An item furnished under production contracts for the purpose of determining compliance with requirements of the specification already established and accepted. It shall be representative of production units though not necessarily made with production line jigs, fixtures and methods.

6.3.2 Production unit.- A unit in its final mechanical form of final production design made by production tools, jigs, fixtures and methods.

CUSTODIAN:

Navy-TD

PREPARING ACTIVITY:

Navy-TD
Project No. 6910-N241

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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Commanding Officer
Naval Training Device Center
Specifications Control Division (3335)
Orlando, Florida 32813

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